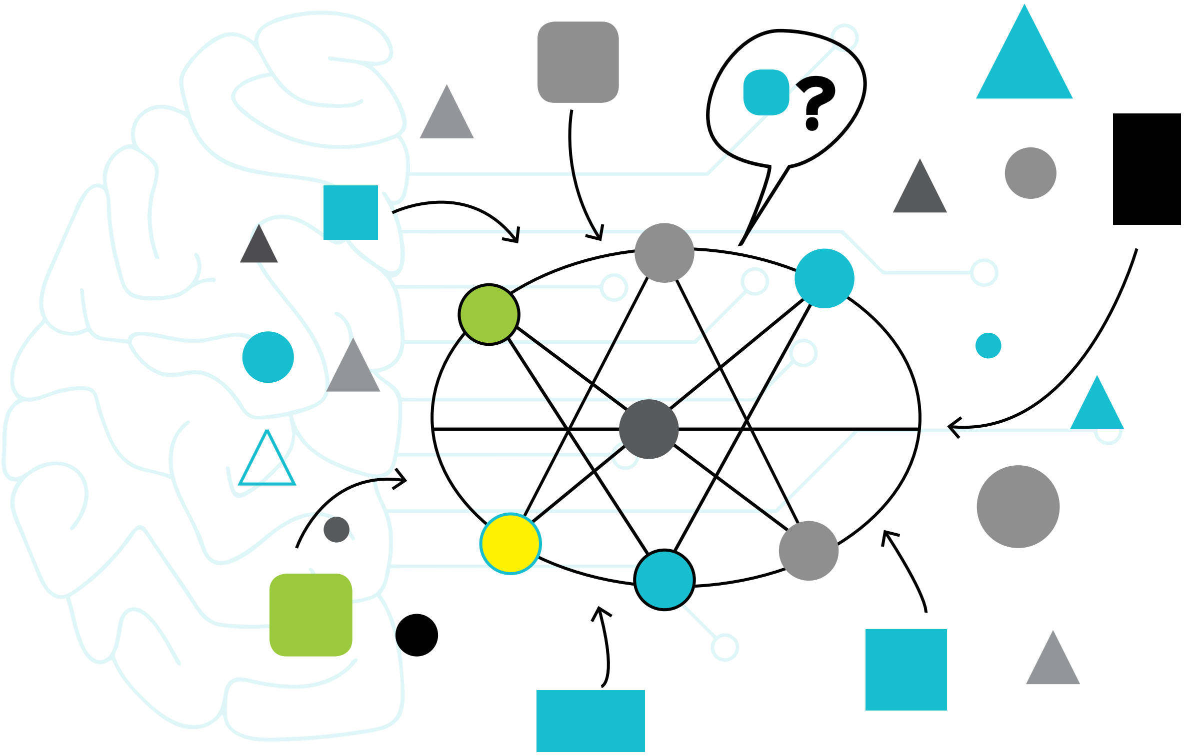
The Long And Short Of FEDERATED MACHINE LEARNING

This article gives a quick insight into what is federated machine learning and why it's popular today



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A cloud refers to a server, public or private, accessible over a network, which is usually the internet. It generally has high processing power and storage, and is suitable for big computations. A cloud can be used to train AI models, but only when the data is available to it. On the other hand, since the cloud is generally a remote system, capturing data directly on it becomes difficult and not feasible. Capturing the data on local devices and transmitting it to the cloud does not always give real-time results. This is where the concept of federated learning comes in.

Federated learning promotes machine learning while the data is on the device. It handles a flexible architecture, which enables a secure process for sensitive data collection and model training. The world currently takes data privacy as an important responsibility. To introduce automation into fields like healthcare, biometrics, etc, real-time sensitive data is the core requirement. The important question therefore is: How do we train a model without collecting sensitive data from the users, storing it, and using it for training? To define optimal training for a machine learning model, relevance of data is an important aspect. Data is best when it comes directly from a source. But